Atsa and the Combos

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

Atsa has just bought the newest Mortal Kombat game. The game has the following set of available buttons: A, B, X, Y, L, R, UP, $DOWN$, $LEFT$, and $RIGHT$.

The game contains N combos, and each combo consists of a specific sequence of buttons that must be pressed in a particular order. For example, one combo could be $DOWN$, $RIGHT$, A, while another could be A, B, X.

Atsa managed to record a sequence of M buttons that were pressed during a match, but he does not know exactly which combos were performed.

A combo could have occurred if its buttons appear consecutively and in the specified order within the recorded sequence. Each button press can belong to at most one combo, meaning that two combos cannot share any positions in the recorded sequence.

Given the list of combos and the recorded sequence of button presses, determine the maximum number of combos that could have been performed.

Input

- The first line contains an integer N ($1 \leq N \leq 1 \times 10^5$).
- Each of the next N lines contains a sequence of buttons representing one combo, with the buttons separated by whitespace.
- The last line contains the sequence of M ($1 \leq M \leq 1 \times 10^6$) buttons recorded by Atsa.

Output

Print a single integer: the maximum number of combos that could have been performed.

Examples

standard input	standard output
3 A B X B X X Y A B X Y	1
5 UP LEFT A A B B X Y X Y UP LEFT UP LEFT A B X Y	3

Note

- The number of total pressed buttons in all combos is at most 5×10^5.